Uday Singh Kanwar

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Education

McMaster University Expected Graduation, April 2027
Bachelor of Engineering in Software Engineering- GPA: 3.9/4.0, Dean's Honour List *Hamilton, Ontario*

Work Experience

AI Engineer Intern May 2025 - Present
Blu Creative- *A software company providing AI-powered web solutions across North America.* *Toronto, ON*

- Building a **Retrieval-Augmented Generation** (RAG)-based conversational system using **GPT-4o, SQLAlchemy**, and **LlamaIndex** to efficiently query documents and a **PostgreSQL** database, improving information access and latency.
- Engineered context-aware query analysis and response generation pipelines via structured **prompt engineering**, enabling accurate and dynamic interaction with **LLMs**.
- Developed and deployed **AI-powered FastAPI** endpoints with real-time context injection from chat and query history, supporting multi-turn conversations and minimizing redundant LLM calls by **70%** through **Redis caching**.

Machine Learning Engineer Jan 2025 - Present
McMaster Aerial Robotics and Drone Team: *Student Club building drones for SUAS Canada competition.* *Hamilton, ON*

- Engineered a **TensorFlo**w object detection **CNN** with **93%** accuracy to enable autonomous drone flight to targets.
- Integrated **NVIDIA Jetson Nano AI Devkit** with drone flight controller to enable real-time, **low-latency** perception to control pipeline, validated in a **Gazebo** simulation.

Software Engineer Intern Jan 2025 - March 2025
Learning Mode AI- *EdTech startup transforming YouTube videos into interactive learning with LLMs.* *Toronto, ON*

- Built **3+ Go** and **Python** based **RESTful** APIs to enable dynamic quiz generation, concise video summaries, and real-time Q&A by leveraging **Redis** and **GPT-4o LLM** integration while working in a **docker** based environment.
- Developed **10+ UI features** and resolved **15+ bugs**, improving responsiveness by **30**% using **React** and **JavaScript**.

Embedded Full-Stack Engineer June 2024 - Oct 2024
Human Endeavour: *Organization advancing senior care with smart and assistive technology* *Vaughan, ON*

- Managed a **MySQL** database handling real-time data from **15+ IoT sensors**, improving query performance by **40**% through index optimization and efficient schema design.
- Built an interactive dashboard using **React.js**, a **Flask** backend, and **Tableau** integration, enabling real-time visualization of **500**+ daily data points, with front-end workflows prototyped in **Figma** for intuitive UI/UX design.
- Developed a **C++** fall detection **algorithm** using an ESP32 micro-controller and accelerometer with **95% accuracy**.
- Built a **trilateration** algorithm with **Ultra-Wideband, AT Commands,** and C++ for indoor tracking at **80%** accuracy.

Technical Projects

AI Travel Agent — LLM-Based Flight Search Assistant | *NVIDIA AIQ Toolkit, Python, NVIDIA NIM, SerpAPI* May 2025

- Developed an AI-powered travel assistant using NVIDIA **AIQ Toolkit**, **ReAct agent** architecture, and **LLaMA-3.3-70B-Instruct** via **NVIDIA NIM** for natural language processing and delivering real-time flight data through a structured integration with the **SerpAPI Google Flights tool**.
- Engineered a **YAML-configured** agent workflow with tool chaining and LLM-guided **slot extraction** for flight details (dates, locations, travel class), enabling accurate, context-aware responses with robust error handling and retry logic.

AWS - Llama LLM Inference Optimization Contest | *Neuron Kernel Interface, Trainium EC2, PyTorch, Python* Apr 2025

- Re-implemented matrix multiplication and scaled dot product attention using **Neuron Kernel Interface** (NKI) to optimize transformer inference on **AWS Trainium EC2 chips.**
- Achieved **1.13×** speedup in **LLaMA inference** throughput and improved accuracy, FLOPs ratio, and performance using **AWS benchmarking** tools.

Credit Card Fraud Detection | *Python, Scikit-learn, Tensorflow, Pandas, NumPy, Matplotlib, Seaborn, Kaggle* Dec 2024

- Evaluated **Logistic Regression, SVM, KNN, and Decision Trees** for fraud detection using **scikit-learn**.
- Achieved 98.7% accuracy (Logistic Regression, SMOTE), 94.2% (Random Under-Sampling), and 92.1% (**TensorFlow Neural Network**) by training the models on an imbalanced credit card transactions dataset.
- Performed data processing by **feature scaling**, **outlier removal**, **correlation analysis**, **data visualization and analytics**.

Hobbies and Clubs | *Google Student Developers' Club, Intramural: Hockey, Cricket, Basketball and Volleyball*

Skills

Languages: Python, Java, Go, JavaScript, C++, C, React.js, Node.js, TypeScript
Technical Proficiencies: NVIDIA AIQ toolkit, NVIDIA NIM, Firebase, Flask, FastAPI, phpMyAdmin, PostgreSQL, TensorFlow, OpenCV, Arduino IDE, MatLab, GitHub, Maven, Docker, Pandas, Numpy, Scikit Learn, Amazon EC2, Matplotlib, JUnit, Pytorch, Software Development Life Cycles (Spiral, Waterfall)